

Menu-Driven Program Using Nested Loops and Switch-Case

Multiplication Tables, Multiplication Grid and Consecutive Integers in C

Aim

To write a menu-driven C program that accepts two positive integers m and n , where $m \leq n$, and performs multiplication-table, multiplication-grid and consecutive-integer operations using loops and a switch statement.

Theory / How the Code Works

The program first reads two positive integers and validates that $m \leq n$. A **do-while loop** repeatedly displays the menu until Exit is selected. A **switch statement** selects the required operation. Simple loops print the two multiplication tables, while nested loops generate the multiplication grid and the groups of consecutive integers. Invalid menu choices are handled by the **default** case.

Menu

Choice	Operation
1	Multiplication tables of m and n
2	Multiplication grid
3	Consecutive integers
4	Exit

Algorithm

1. Start.
2. Read two integers m and n .
3. If $m \leq 0$, $n \leq 0$, or $m > n$, display an error and repeat the input.
4. Display the four-option menu and read choice.
5. Use `switch(choice)` to perform the selected operation.
6. For choice 1, run i from 1 to 10 and print $m \times i$ and $n \times i$ side by side.
7. For choice 2, use nested loops: rows run from 1 to m and columns from 1 to n ; print $i \times j$ in each grid position.
8. For choice 3, run i from m to n . For each i , run j from 0 to 5 and print $i + j$.
9. For choice 4, display an exit message.
10. For any other choice, display an invalid menu-choice message.
11. Repeat the menu until choice 4 is selected.
12. Stop.

Test-Case Table

Test	Input	Choice	Expected result
1	$m=3, n=5$	1	Tables of 3 and 5 printed side by side from $x1$ to $x10$.
2	$m=3, n=5$	2	Grid with rows 1-3 and columns 1-5 is displayed.
3	$m=3, n=5$	3	For 3, 4 and 5, each number and its next five integers are printed.
4	$m=3, n=5$	4	Program exits normally.

Test	Input	Choice	Expected result
5	m=-2, n=5	-	Input is rejected because m is not positive.
6	m=6, n=3	-	Input is rejected because m > n.
7	m=1, n=1	1	Multiplication table of 1 is displayed in both columns.
8	m=3, n=5	8	Invalid menu-choice message; menu is displayed again.

C Program

```
#include <stdio.h>

int main()
{
    int m, n, choice;
    int i, j;

    do
    {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n);

        if (m <= 0 || n <= 0 || m > n)
            printf("Invalid input! Enter positive integers with m <= n.\n");
    } while (m <= 0 || n <= 0 || m > n);

    do
    {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n",
                    m, n, m, n);

                for (i = 1; i <= 10; i++)
                {
                    printf("%d x %d = %-10d", m, i, m * i);
                    printf("%d x %d = %d\n", n, i, n * i);
                }
                break;

            case 2:
                printf("\nMultiplication Grid\n\n");

                printf("    ");
                for (j = 1; j <= n; j++)
                    printf("%4d", j);

                printf("\n");

                for (i = 1; i <= m; i++)
                {
                    printf("%4d", i);

                    for (j = 1; j <= n; j++)
                        printf("%4d", i * j);

                    printf("\n");
                }
                break;

            case 3:
                printf("\nConsecutive Integers\n");

                for (i = m; i <= n; i++)
                {
                    printf("%d: ", i);

                    for (j = 0; j <= 5; j++)
                        printf("%d ", i + j);

                    printf("\n");
                }
                break;

            case 4:
                printf("\nExiting the program...\n");
                break;

            default:
```

```
        printf("\nInvalid menu choice! Please enter 1, 2, 3 or 4.\n");
    }
} while (choice != 4);
return 0;
}
```

Compilation and Execution

The program was compiled using `gcc -Wall -Wextra menu_multiplication.c -o menu_multiplication`. The compilation completed successfully. It was then executed with `m = 3` and `n = 5`, testing menu choices 1, 2, 3, an invalid choice 8, and finally Exit.

Sample Execution

Enter two positive integers m and n ($m \leq n$):

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Multiplication Tables of 3 and 5 ($m=3$, $n=5$)

3 x 1 = 3	5 x 1 = 5
3 x 2 = 6	5 x 2 = 10
3 x 3 = 9	5 x 3 = 15
3 x 4 = 12	5 x 4 = 20
3 x 5 = 15	5 x 5 = 25
3 x 6 = 18	5 x 6 = 30
3 x 7 = 21	5 x 7 = 35
3 x 8 = 24	5 x 8 = 40
3 x 9 = 27	5 x 9 = 45
3 x 10 = 30	5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Multiplication Grid

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Consecutive Integers

3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Invalid menu choice! Please enter 1, 2, 3 or 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Exiting the program...

Result

The menu-driven C program was written, compiled and executed successfully. It correctly prints the multiplication tables of m and n side by side, generates the multiplication grid, prints consecutive integers, handles invalid menu choices, validates m and n, and continues until Exit is selected.